

BFW STEAM HEATER VER 1

Operator Training Simulator · E-101

Chulalongkorn University · Department of Chemical Engineering

DOCUMENT NO. : BFWH-V1-IN-CE-001

DOCUMENT TITLE : CAUSE AND EFFECT DIAGRAM

Teaching document generated from the simulator logic (bmsTrip, tripConditions, alarm list).

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
R0	03-Oct-26	Issued for teaching	A. Soottitantawat	—	—	

BFW STEAM HEATER VER 1 · E-101 CAUSE and EFFECT DIAGRAM Chulalongkorn University · Department of Chemical Engineering								Client :		Chulalongkorn University · Department of Chemical Engineering					REV	DATE	ORIGINATOR	CHECKER	APPROVER
								PROJECT :		BFW Steam Heater Ver 1 · OTS					R0	03-Oct-26	A. Soottitantawat	—	—
								DOCUMENT No. :		BFWH-V1-IN-CE-001									
								DOCUMENT TITLE :		Cause and Effect Diagram									
								P&ID :		P&ID-001 rev C / BFWH-V1-PID-001 rev D									SHEET 1 of 1
Causes								EFFECTS											
								XV-100 · Steam shut-off valve (SIS)	FV-104 · Steam control valve (FIC-104 → MAN 0 %)	TIC-101 · Outlet temperature controller → MAN	SIS · SIS I-100 state	1ST · First-out annunciation	AMB · Flashing lamp (amber)	RED · Flashing lamp (red)	AUD · Audible alarm				
Trip Action								C	C	MAN	TP/RS	F	A	A	A	Notes: 1. Trip action codes: C = close; MAN = controller forced to manual; TP = SIS tripped (manual reset required); RS = SIS stays in IDLE; F = first-out; A = activate. 2. FSLL-101 is armed only when the SIS is in RUN (XV-100 open); the other trips are always armed. 3. Trip effects latch until reset. Reset only when no TSHH-108 / PSHH-100 cause remains; the SIS then returns to IDLE and XV-100 is opened again by the operator. 4. Manual trip; immediate, no delay. Same effects as any SIS trip. 5. Alarm active only with BFW flowing or with steam on, as in the simulator alarm list (suppression by state). 6. TIAH-108 and TSHH-108 share TT-108, which is independent of the control transmitter TT-101 (LOPA tab: own element TE-108B recommended for the alarm — HOLD). 7. PI-100 alarm, PSHH-100 trip and the indicator share PT-100 (Safety layers tab: own transmitter recommended — HOLD). FSLL-101 shares FT-101 with FIC-101 in the same way. 8. Start permissives: SIS reset (IDLE), FIC-101 ≥ 50 m³/h, FIC-104 in MAN with output ≤ 10 %, trap block valves V-101 and V-102 open. Not a trip: no valve moves. 9. BFW pumps P-101A/B are not tripped: water keeps flowing through the tubes after a trip (FV-101A fails open).			
Item No.	Tag No.	Tag Service Description	Trip/Alarm-Direction	Setting	Eng. Unit	Delay	Note no.												
001	HS-ESD	Emergency shutdown push-button (DCS / SIS screen)	Tripped (opened contact)	—	—	0 s	4	X	X	X	TP	X		X	X				
002	FIC-101	BFW flow to E-101 low alarm (FT-101)	Low	80	m³/h		5						X		X				
003	FIC-101	BFW flow to E-101 high alarm (FT-101)	High	120	m³/h		5						X		X				
004	FSLL-101	BFW flow low-low trip (FT-101)	Low-Low	30	m³/h	5 s	2, 5	X	X	X	TP	X		X	X				
005	TI-103	BFW inlet temperature low alarm	Low	12	°C		5						X		X				
006	TI-103	BFW inlet temperature high alarm	High	25	°C		5						X		X				
007	TIC-101	BFW outlet temperature low alarm (TT-101)	Low	67	°C		5						X		X				
008	TIC-101	BFW outlet temperature high alarm (TT-101)	High	74	°C		5						X		X				
009	TIAH-108	BFW outlet high alarm (independent TT-108)	High	85	°C		6						X		X				
010	TSHH-108	BFW outlet high-high trip (TT-108)	High-High	95	°C	5 s	6	X	X	X	TP	X		X	X				
011	PI-100	E-101 shell pressure high alarm (PT-100)	High	200	kPa g		7						X		X				
012	PSHH-100	E-101 shell pressure high-high trip (PT-100)	High-High	250	kPa g	3 s	7	X	X	X	TP	X		X	X				
013	PI-105	Flash steam header pressure low alarm	Low	150	kPa g								X		X				
014	PI-105	Flash steam header pressure high alarm	High	240	kPa g								X		X				
015	FIC-104	Flash steam flow high alarm (FT-104)	High	13000	kg/h		5						X		X				
016	PDI-109	Tube-side differential pressure high alarm	High	80	kPa		5						X		X				
017	I-100	Open XV-100 refused: start permissive not met	Not met	—	—		8				RS		X						
018																			
019																			